

PRIYANKA SIWACH

New York, NY • (929) 582-0786 • priyankasiwach214@gmail.com • github.com/PriyankaSiwach • linkedin.com/in/priyanka-siwach

EDUCATION

Queens College, City University of New York (CUNY)

New York, NY

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- GPA: 3.75/4.00, Dean's List, Honors Student
- Relevant Coursework: Data Structures & Algorithms, OOP, Cloud Computing, AI/LLM Systems, Database Systems

EXPERIENCE

AIN Ventures

New York, NY

Software Engineer Intern

June 2026 – August 2026

- Built a **Python-based sourcing agent** in Brain, AIN Ventures' internal AI platform, to re-surface promising founders below the 90-point auto-reject threshold, adding **5+ founders per day** to the outreach pipeline.
- Built a **200+ case adversarial test suite in Python** to stress-test AI guardrails, identify failure cases, and validate fixes across edge-case and malicious inputs.

Museum of Modern Art (MoMA)

New York, NY

Cybersecurity Engineer Intern

March 2024 – May 2024

- Updated and **validated identity**, photo, and badge-access records for **500+ MoMA employees**, strengthening the accuracy and integrity of the museum's physical access-control system.

PROJECTS

MiniLSM — Crash-Safe Key-Value Storage Engine (Python, Bloom Filter, Compaction, WAL)

- Achieved **zero lost acknowledged writes** across **1,000 SIGKILL** trials (**~2.2M ops**; test caught an injected bug in **20/20** runs), by building a checksummed **write-ahead log** with torn-write recovery and atomic renames.
- Cut disk reads per missing-key lookup **99%** ($21.96 \rightarrow 0.19$) and raised throughput **4.5x** ($5.7K \rightarrow 25.4K/sec$), by adding per-file **Bloom filters** (10 bits/key, **0.85%** measured false-positive rate, ~25 KB per 20,000 keys).
- Cut disk usage **54%** ($4.9 \rightarrow 2.2$ MB) and missing-key p50 latency **~40x** ($75 \rightarrow 1.9 \mu s$) on a seeded, overwrite-heavy benchmark, by writing streaming **k-way merge compaction** (**44 SSTables** $\rightarrow$ **1**; write amplification **1.62**).

ChefCoach — AI Recipe Scanner App (React, TypeScript, Supabase, OpenAI)

- Avoided **905 of 1,000 OpenAI calls (90.5%)** in a simulated request benchmark, cutting billable API usage, by building an **LRU cache** and a **token-bucket rate limiter** in front of the GPT-4o vision API.
- Kept recipe scanning working through OpenAI API outages in an **App Store-published** iOS app by designing a local fallback path that takes over when the API fails.

Applyfy — Job Prep & Resume Tailoring Website (applyfy.net) (Next.js, TypeScript, Supabase, OpenAI, Stripe, AWS)

- Built Applyfy, an AI job platform integrating OpenAI GPT-4o across **12 features**, protected by Redis-backed rate limiting (**5 req/min/user**), reaching **36 signups**.
- Engineered a serverless AWS resume-parsing pipeline (API Gateway, Lambda, S3, DynamoDB, WAF, CDK); diagnosed and isolated two independent throttling layers (WAF edge rate limiting vs. Lambda account concurrency) under **50-request concurrent load**, sustaining **45 req/sec** at **1.9s latency** for legitimate traffic.

Collaborative Whiteboard — Real-Time Multi-User Sync (Next.js, Node.js, WebSockets, Yjs CRDTs, Redis Pub/Sub)

- Sustained **100 concurrent WebSocket** connections at **100%** connection success in **k6** load testing, by building real-time collaborative sync with **Yjs CRDTs** and **Redis Pub/Sub** to fan out document updates across multiple servers.

CERTIFICATIONS

- **AWS Certified Solutions Architect — Associate**, Amazon Web Services (AWS) — Issued June 2026
- **AWS Certified Cloud Practitioner**, Amazon Web Services (AWS) — Issued November 2025

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript

Frameworks & Libraries: React, Next.js, FastAPI, Node.js

Databases & Cloud: Redis, PostgreSQL, AWS (EC2, S3, Lambda, API Gateway, DynamoDB)

Systems Concepts: WAL, LSM-trees, Bloom filters, compaction, LRU caching, rate limiting, CRDTs, Pub/Sub

Tools: Git, GitHub Actions, pytest, Docker